

DSBD 4

Code:

```
DROP DATABASE IF EXISTS bookflow_db;
CREATE DATABASE bookflow_db;
USE bookflow_db;

CREATE TABLE Books (
    book_id INT AUTO_INCREMENT PRIMARY KEY,
    title VARCHAR(200) NOT NULL,
    author VARCHAR(150) NOT NULL,
    genre VARCHAR(100),
    description TEXT,
    published_year INT
);

INSERT INTO Books (title, author, genre, description, published_year) VALUES
('The Martian', 'Andy Weir', 'Science Fiction', 'An astronaut struggles to survive alone on Mars.', 2011),
('Dune', 'Frank Herbert', 'Science Fiction', 'Epic desert planet adventure.', 1965),
('The Hitchhikers Guide to the Galaxy', 'Douglas Adams', 'Science Fiction', 'Comedy space adventure.', 1979),
('Pride and Prejudice', 'Jane Austen', 'Romance', 'Classic romance novel.', 1813),
('The Great Gatsby', 'F. Scott Fitzgerald', 'Classic', 'American dream story.', 1925),
('Enders Game', 'Orson Scott Card', 'Science Fiction', 'Young military genius in space.', 1985),
('Foundation', 'Isaac Asimov', 'Science Fiction', 'Galactic empire saga.', 1951),
('Neuromancer', 'William Gibson', 'Cyberpunk', 'Cyberpunk hacking adventure.', 1984),
('Brave New World', 'Aldous Huxley', 'Dystopian', 'Future controlled society.', 1932),
('Fahrenheit 451', 'Ray Bradbury', 'Dystopian', 'Books are outlawed.', 1953);

ALTER TABLE Books
    ADD COLUMN embedding JSON;

UPDATE Books SET embedding = JSON_ARRAY(0.90,0.80,0.20) WHERE book_id=1;
UPDATE Books SET embedding = JSON_ARRAY(0.85,0.75,0.25) WHERE book_id=2;
UPDATE Books SET embedding = JSON_ARRAY(0.80,0.70,0.30) WHERE book_id=3;
UPDATE Books SET embedding = JSON_ARRAY(0.10,0.20,0.90) WHERE book_id=4;
UPDATE Books SET embedding = JSON_ARRAY(0.15,0.30,0.80) WHERE book_id=5;
UPDATE Books SET embedding = JSON_ARRAY(0.88,0.78,0.22) WHERE book_id=6;
UPDATE Books SET embedding = JSON_ARRAY(0.83,0.73,0.27) WHERE book_id=7;
UPDATE Books SET embedding = JSON_ARRAY(0.75,0.65,0.35) WHERE book_id=8;
UPDATE Books SET embedding = JSON_ARRAY(0.20,0.25,0.85) WHERE book_id=9;
```

```
UPDATE Books SET embedding = JSON_ARRAY(0.18,0.28,0.82) WHERE book_id=10;
```

```
SELECT book_id,title,embedding
FROM Books
ORDER BY book_id;
```

```
SELECT
  book_id,
  title,
  author,
  genre,
  ROUND(
    SQRT(
      POW(CAST(JSON_EXTRACT(embedding,'$[0]') AS DECIMAL(10,4))-0.90,2)+
      POW(CAST(JSON_EXTRACT(embedding,'$[1]') AS DECIMAL(10,4))-0.80,2)+
      POW(CAST(JSON_EXTRACT(embedding,'$[2]') AS DECIMAL(10,4))-0.20,2)
    ),4) AS l2_distance
FROM Books
ORDER BY l2_distance ASC
LIMIT 3;
;
```

Output:

book_id	title	author	genre	
1	1 The Martian	Andy Weir	Science Fiction	
2	6 Enders Game	Orson Scott Card	Science Fiction	
3	2 Dune	Frank Herbert	Science Fiction	

book_id	title	embedding
1	1 The Martian	[0.90, 0.80, 0.20]
2	2 Dune	[0.85, 0.75, 0.25]
3	3 The Hitchhikers Guide to the Galaxy	[0.80, 0.70, 0.30]
4	4 Pride and Prejudice	[0.10, 0.20, 0.90]
5	5 The Great Gatsby	[0.15, 0.30, 0.80]
6	6 Enders Game	[0.88, 0.78, 0.22]
7	7 Foundation	[0.83, 0.73, 0.27]
8	8 Neuromancer	[0.75, 0.65, 0.35]
9	9 Brave New World	[0.20, 0.25, 0.85]
10	10 Fahrenheit 451	[0.18, 0.28, 0.82]